

Theft, Trade, and Social Order

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1 Abstract

In the absence of institutions that protect property rights, what conditions allow trade to emerge among self-interested individuals capable of violence? This paper studies a production economy in which individuals can trade peacefully or appropriate goods through conflict. Because conflict, not autarky, is the outside option to trade, peaceful exchange must compete with an alternative that is not equally unattractive to everyone: theft rewards whoever prevails. Trade emerges only when the distribution of economic goods happens to match the distribution of coercive power, and decentralized economic activity provides no mechanism ensuring that alignment.

Organizing into groups could raise returns to specialization in production or coercion, but any division of a group's winnings is itself contested: a member participates only if the group offers more than what she could take by fighting alone. Because coercive power determines each player's claim on cooperative gains,

every player has an incentive to expand her own power at the group's expense, undermining the organization specialization would otherwise reward.

A transfer schedule that compensates each player according to her coercive power can, in principle, make peaceful exchange as rewarding as theft. But the existence of such a schedule establishes only the feasibility, not the implementation, of peace: absent an institution to enforce it, each player has an incentive to alter her own coercive power to improve her post-transfer position, unraveling the transfer for the same reason organization unravels. Sustained peace, this paper argues, requires an enforcement mechanism independent of the coercive power it must restrain — a condition decentralized exchange among self-interested individuals cannot supply on its own.

2 Introduction

2.1 Motivation and Related Literature

Barry Weingast noted a fundamental political paradox: “A government strong enough to protect property rights and enforce contracts is also strong enough to confiscate the wealth of its citizens” (Weingast 1995). The roots of this political dilemma lie in behavior at the individual level. Exchange between individuals is fundamental to the creation of value, yet the production and accumulation of valuable goods also create incentives for appropriation. The national-level problem identified by Weingast is thus a manifestation of a more fundamental individual problem: the capacity to create gains from exchange also creates the opportunity to take those gains by force.

These dual dilemmas raise a fundamental question: under what conditions will individuals capable of both exchange and appropriation choose peaceful trade? Trade theory traditionally answers a related question by comparing trade to autarky: individuals specialize and exchange because withdrawing from exchange entirely is worse for everyone. This paper replaces the autarky outside option with conflict, and that replacement is not innocuous. Autarky is worse for everyone; conflict is not — conflict rewards whoever prevails. As a result, peaceful trade must compete with the coercive alternative that some individuals may prefer. The central dilemma only disappears when the distribution of economic goods produced by trade matches the distribution that coercive power would produce. Because nothing in decentralized economic activity ensures that alignment, the conditions under which self-interested individuals choose trade over conflict are considerably narrower than the classical case, and the central question of this paper is precisely when — and by what means — that alignment can be achieved.

Institutions emerged to resolve this dilemma at the societal level, either by restraining violence directly or by shaping the distribution of goods so that peaceful exchange competes successfully with the coercive alternative. In doing so, the emergence of institutions replaces the individual dilemma with a societal dilemma. Yet

the origins of these institutions predate the written record, and may be permanently unrecoverable (Willhoite Jr 1976; Fukuyama 2011).¹ Rather than reconstruct this history, this paper asks whether institutions are the only mechanism capable of resolving this dilemma, and, if so, why. A large literature has examined institutions and the connected problems of appropriation and social order. This paper builds directly on that work, beginning with the literature that studies the individual production and appropriation decisions from which those problems arise.

Skogh and Stuart (1982), Skaperdas (1992), De Meza and Gould (1992), Hirshleifer (1995), Grossman and Kim (1996), Grossman (2002), Bates, Greif, and Singh (2002), and Baker, Bulte, and Weisdorf (2010) all study environments in which individuals must allocate scarce resources between producing goods and protecting or appropriating them. In these models, however, conflict defines the strategic environment in which allocation decisions are made. This paper makes the form of interaction itself a choice. Individuals who are capable of appropriation may instead exchange their production through trade, allowing the model to ask not only how the possibility of predation affects production and protection, but under what conditions self-interested individuals choose peaceful exchange rather than conflict.

Polachek (1980) established a literature in which agents choose between trade and conflict, although his analysis concerns trade between nations rather than exchange among individuals, as does the later work of Martin, Mayer, and Thoenig (2008). Within the literature directly comparing trade and conflict, the closest antecedents to the present model are Skaperdas and Syropoulos (Skaperdas and Syropoulos 2001, 2002), who combine production, trade, and conflict to examine how insecure property rights affect the gains from trade and incentives to invest in security. This paper turns the question around. Rather than asking how insecurity affects trade, it asks when trade can emerge in an economy in which appropriation remains an available alternative. The existence of gains from trade is not sufficient: peaceful exchange requires that the distribution of those gains leaves every participant better off than exercising the violent alternative.

The problem of reconciling economic activity with the capacity for violence is also central to the institutional literature. While the models discussed above explore the individual decisions over production, appropriation, and exchange, institutional scholars have approached the same underlying problem by studying the organizations and rules through which societies constrain violence and sustain cooperation.

The institutional literature has long connected political organization to the control of coercion and the distribution of economic resources. Max Weber (Weber 1946) described forms of political organization and

¹Evidence for record-keeping and territorial markers extends into deep prehistory [schmandt1978]. Early inter-group and inter-species violence has left archaeological traces [crevecœur2021; meijer2024], though its causes and extent remain debated [rutar2023]. Comparative evidence from non-human primates suggests the capacity to constrain intragroup violence may predate modern humans entirely [masters1991].

canonically placed control over legitimate violence at the center of the modern state. Mancur Olson connected coercive capacity to both collective action and political organization: selective incentives help overcome the collective-action problem (Olson Jr 1971), while the stationary-bandit metaphor illustrates how an actor capable of predation can acquire an incentive to establish order rather than continually appropriate output (Olson 1993). Avner Greif examines the origins of various historical institutions capable of constraining violent behavior and supporting exchange (Greif 2006). Together, these authors explain how institutions constrain violence and sustain social order. This literature, however, generally takes the existence of such institutions as a starting point. The current paper instead treats their existence as an open question rather than a premise.

North, Wallis, and Weingast (NWW), with additional contributors, place the control of violence at the center of their explanation of social order (North, Summerhill, and Weingast 2000; North, Wallis, and Weingast 2009; North et al. 2013; Wallis 2026). Individuals with the capacity for violence form organizations that control resources and threaten violence against others. Social order depends upon these organizations having the capacity to capture the benefits of economic activity so that powerful actors prefer cooperation to violence. This logic produces the proportionality principle (Cox, North, and Weingast 2019): stable arrangements must distribute benefits in relation to the violence potential of their participants. Organizations are therefore both a means of exercising coercive power and a means of organizing cooperation among those capable of violence. Skaperdas and Syropoulos were the first to publish a form of the proportionality principle as a result of the tension between coercive power and peaceful exchange (Skaperdas and Syropoulos 1997). The current paper identifies the exact result of Skaperdas and Syropoulos and extends it to a condition for exchange to exist at all.

Van Bavel, Ansink, and Van Besouw formalize important elements of the NWW framework by beginning with distinctions between elites and other actors and examining production and appropriation within Limited Access and Open Access Orders (Van Besouw, Ansink, and Bavel 2016; Van Bavel, Ansink, and Van Besouw 2017). Acemoglu and Robinson (2006) model voting, transfers, wealth distribution, revolutions, coups, and similar important institutions directly.

This paper approaches institutional development from another angle. Beginning with Ricardian trade, I empower agents to produce and protect, appropriate and exchange goods without assuming any exogenous institutions that protect property rights. However, because a market is an institution, players can endogenously choose to participate in a market institution defined by the following properties:

- Consumer goods are fungible within each type and freely transferable within a market

- All players in a market are price takers
- Each participant's spending cannot exceed the market value of their production
- Market clearing
- Exchange requires two-way consent

The gains from specialization that sustain a market do not extend to security because participating in the market means accepting its property rights. Individuals who respect property rights value security only for deterrence, not for appropriation. Organizing attempts to extend the gains from specialization to security as well — pooling coercive capacity the way a market pools consumer goods. I show that no self-enforcing division of the resulting gains can be sustained: the same dilemma that constrains peaceful exchange between individuals reappears, unresolved, within the organizations meant to overcome it.

The same dilemma appears at the level of the whole economy. I show that a schedule of transfers exists that can align the distribution generated by competitive exchange with the distribution that coercive power would produce. Trade occurs, and violence does not — but coercion has not disappeared; it is preserved in the transfers themselves. The existence of this schedule establishes only the feasibility, not the implementation, of economy-wide peace.

The transfer schedule is organization taken to its limit — a coalition spanning the entire economy — and it fails for exactly the reason organization does at any smaller scale. The market, extended by transfers, is one institution self-interested individuals might choose, and it is not sufficient to sustain peaceful exchange without an institution to enforce it — one with the same power to confiscate the distribution it is meant to protect. This is consistent with Weingast's observation: every institution we currently have falls short of resolving the same dilemma at the societal level.

2.2 Classical Political Theory

To develop modeling assumptions relevant to a world without externally enforced property rights, I turn to the intellectual tradition that explicitly considers such a world. Classical political philosophers debated the origins of justice, society, and individual rights; economists have generally taken the institutions that define and enforce those rights as given.

I begin by assuming that individuals are self-interested and rational: they pursue their own ends and choose actions they believe will best achieve them. These are basic economic assumptions, but they are grounded in

the philosophy of Machiavelli, Hobbes, and Hume. Machiavelli especially analyzes human action as revealed by what people do rather than by what society thinks they should do (Machiavelli 2003). Hobbes ignores the social pressures and assumes that individuals are motivated by their own desires. (Hobbes 2017). Hume separates ends from means: people pursue their desires and employ their higher faculties to better acquire the things they want rather than to determine what it is they should want (Hume 2016). In this sense, self-interest determines the objectives and rationality governs the methods.

While Locke might agree with Hobbes that “the voluntary actions, and inclinations of all men, tend, not only to the procuring, but also to the assuring of a contented life,” (Hobbes 2017), Locke constrains the pursuit of those ends by natural law. Individuals have a natural entitlement to their own labor and, through that labor, may acquire property in what they produce (Locke 2018). Grotius reaches a similar conclusion with an important qualification: production establishes ownership only when the other inputs to production do not already belong to another (Grotius 2024). Thus, for both Locke and Grotius, individuals may acquire and defend a property claim through their own productive activity even in the absence of institutions backing that claim.

The inevitability of conflict in Hobbes’s “Leviathan” arises because some people will not respect that claim to a property right without a central authority. When one can never be certain who might attempt theft, Hobbes asserts that people ‘anticipate’ that theft and act with preemptive violence. Hume similarly argues that without the institutions protecting property, individuals “wou’d run head-long into every kind of injustice and violence” (Hume 2016). Their solutions differ, but both identify the insecurity of possession as a source of conflict. Both Locke and Grotius require consent for the legitimate transfer of an individual’s possessions; taking those possessions without consent provides just grounds for violence in response. It is that reactive violence that constrains theft in Locke’s natural state without institutions. Any encroachment on one’s natural rights could be met with escalated violence, for “men who judge their own case will always [...] overestimate the amount of harm they have suffered” (Nozick and Laqueur 1971). The injured party may gather others to enforce the natural law, bringing sufficient power to bear against the transgressor.

The market institution defined in the preceding section embeds a particular form of property rights: participants recognize claims to the goods they produce and acquire through consensual exchange, and transfers require mutual consent. Players choose whether to respect these claims through their actions. Trade respects ownership by requiring consent; theft disregards the owner’s claim by attempting appropriation without consent. The model therefore specifies the property rules governing market exchange without assuming that all players respect them.

The power individuals bring to conflict depends on their capabilities. Hobbes, like Rousseau, recognizes

innate differences in those capabilities. Rousseau explicitly assumes inequalities “established by nature,” including differences in health, physical strength, and qualities of mind. (Rousseau 2004) Unlike Hobbes, Locke and Hume, Rousseau does not require these natural differences to be small. Hobbes claims that “the weakest has strength enough to kill the strongest”. I synthesize those viewpoints; although individual capabilities are not constrained in my model, conflict is treated as a stochastic contest; differences in capability affect the probability of success rather than determine its outcome.

Machiavelli counsels his prince to train himself and his armies in the arts of war through study and action, even more while at peace than during a war. Individual capabilities can be improved through effort. The capacity for violence is an endogenous choice, but not a costless one. Montesquieu explicitly shows that commerce is the opportunity cost of military power. I incorporate these insights into my model. Individuals can produce ‘security’ through labor, but that labor is constrained through a limited endowment. Any labor spent developing security can not be used to produce the things the individual desires.

While individuals can produce each of the goods they desire, producing all of those goods individually is not efficient. Plato begins his account of the origin of society from precisely this observation: “each of us is not self-sufficient but is in need of much” (Plato 2000). Because individuals have different needs and different productive abilities, specialization allows them to satisfy those needs through the production of goods for others. I incorporate the same insight by assuming that individuals value a variety of goods. This creates potential gains from specialization and exchange and, consequently, a peaceful alternative to acquiring the production of others through appropriation. The model asks which alternative individuals choose when both appropriation and exchange are available actions.

3 Model of Conflict and Trade

3.0.1 Game

The ‘Theft and Trade’ model is a class of games that extends Ricardian trade (Ricardo 1888) by incorporating the assumptions developed in the previous section. A theft and trade game is a dynamic two-stage game with complete and imperfect information. Formally, the game

$$\Gamma = (\mathcal{N}, \Sigma, \pi) \tag{1}$$

is defined with the normal objects: the set of players, the strategy profile space and a payoff function mapping

a strategy profile to a payoff profile. After a brief overview of the game, I explain each of these objects in detail.

In the first round, players produce M consumer goods, and one *security good*. The security good determines a player's coercive power, which influences the likelihood a player can successfully appropriate goods from other players or defend goods from appropriation. In the second round, players choose how they would like to interact with each other player. These round-one production and round-two interaction choices govern the allocation of goods among players, and players derive utility from their consumption.



Figure 1: Game flow

3.0.2 Players

$\mathcal{N}$ is the set of players.

$$\mathcal{N} = \{1, 2, \dots, N\} \quad (2)$$

The players are self-interested and rational. They maximize expected utility. When participating in a market, players are price-takers, but otherwise players behave strategically.

Players have a common utility function U . Utility is C^2 , increasing, strictly quasi-concave, homogenous of degree one, and normalized so $U(\mathbf{0}) = 0$. All consumer goods are essential.

3.0.3 Round One

There are M types of fungible, transferable consumer goods in the economy and one nontransferable *security good*. The set of consumer goods is

$$\mathcal{M} = \{1, 2, \dots, M\} \quad (3)$$

S_n is a player's security good, sometimes called *power*. Nature endows each player with a set of production functions and a fixed quantity of labor $\tau > 0$. These endowments are common knowledge. Formally, the production functions for player n are:

$$f_{nm} : \mathbb{R}^+ \rightarrow \mathbb{R}^+ \forall m \in \mathcal{M}, \quad f_{nS} : \mathbb{R}^+ \rightarrow \mathbb{R}^+. \quad (4)$$

Each production function $f_{nm}, f_{nS} \in C^2$ is twice continuously differentiable, increasing and strictly concave with $f_{nm}(0) = f_{nS}(0) = 0$. The production functions take labor as the sole input; there are no intermediate or capital goods in this economy. Each player has an identical labor endowment τ^2 measured in units of time. In round one, players simultaneously allocate their labor endowments across production activities with the constraint

$$s_n + \sum_{m=1}^M t_{nm} = \tau \quad \forall n \in \mathcal{N}. \quad (5)$$

These allocations cannot be revised once production begins. Production is complete before round two begins at which point each player's labor allocation become common knowledge. L_n is player n 's labor allocation in round one.

$$L_n = (t_{n1}, t_{n2}, \dots, t_{nM}, s_n) \quad (6)$$

The vector of labor allocations is a labor allocation profile L . The space of labor allocation profiles forms the domain of a player's contingent action plan in round two.

$$L = (L_1, L_2, \dots, L_N) \quad L \in \mathcal{L} \quad \mathcal{L} = \mathbb{R}^{N \times (M+1)} \quad (7)$$

X_n^P is player n 's production bundle of consumer goods. X^P is the profile of production bundles.

$$X_n^P = (f_{n1}(t_{n1}), \dots, f_{nM}(t_{nM})) \quad (8)$$

$$X^P = (X_1^P, X_2^P, \dots, X_N^P) \quad X^P \in \mathbb{R}_+^{N \times M} \quad (9)$$

²I impose identical labor endowments for simplicity and parsimony. Allowing heterogeneous labor endowments does not alter the substantive results.

3.0.4 Round Two

In round two, each player simultaneously chooses a *posture* towards each other player based on the collective production choices from round one³.

Player n chooses posture $a_{n,j}$ towards player j .

$$a_{n,j} \in \mathcal{A}_{n,j} = \{\text{none, theft, trade}\} \quad n \neq j \quad (10)$$

Player n 's action is a $N - 1$ vector of postures.

$$a_n = (a_{n,1}, \dots, a_{n,n-1}, a_{n,n+1}, \dots, a_{n,N}) \quad a_n \in \mathcal{A}_n = \prod_{j \neq n} \mathcal{A}_{n,j} \quad (11)$$

The function ρ_n maps a labor allocation profile to player n 's action.

$$\rho_n : \mathcal{L} \rightarrow \mathcal{A}_n \quad a_n = \rho_n(L) \quad (12)$$

Thus, an action profile is the vector of player actions.

$$a = (a_1, \dots, a_N) \quad A \in \mathcal{A} = \prod_{i=1}^N \mathcal{A}_i \quad (13)$$

3.0.5 Strategies and Strategy Profiles

Player n 's strategy $\sigma_n \in \Sigma_n$ is an ordered pair consisting of player n 's labor allocation in round one and the function ρ_n that forms a contingent plan for any labor allocation profile.

$$\sigma_n = (\mathbf{L}_n, \rho_n) \in \Sigma_n, \quad (14)$$

A strategy profile σ is

$$\sigma = (\sigma_1, \sigma_2, \dots, \sigma_N) \in \Sigma, \quad \Sigma = \prod_{i=1}^N \Sigma_i \quad (15)$$

³This choice is termed a posture because it is the interaction the player desires. However, the interaction between players that is realized depends on the postures of both players.

The payoff function π is a composite function that maps strategy profiles to payoff profiles. The components of π are an outcome function g that maps strategy profiles to a joint lottery over consumption bundle allocations and a value function v that assigns players their expected utility of the lottery.

$$\pi : \Sigma \rightarrow \mathcal{R}_+^N \quad \pi \equiv v \circ g \quad g : \Sigma \rightarrow \Delta((\mathbb{R}_+^M)^N) \quad v : \Delta((\mathbb{R}_+^M)^N) \rightarrow \mathbb{R}_+^N \quad (16)$$

The strategy profile induces a partition on the set of players. The outcome function defines independent lotteries for sets in the partition. The remainder of Section section 3 explains how the strategy profile partitions the set of players, and how the outcome function creates lotteries for the players.

3.0.6 Interaction Graph

An action profile induces an undirected interaction graph. For each pair of players $i, j \in \mathcal{N}$, $i < j$, their postures toward one another assign the edge $e_{ij} = \{i, j\}$ to an interaction edge set. If either player adopts a ‘theft’ posture, the pair are assigned an edge in the set of Conflict edges.

$$E^C = \{e_{ij} : a_{i,j} = \text{theft or } a_{j,i} = \text{theft} \quad i, j \in \mathcal{N}, i < j\} \quad (17)$$

Conflict and Trade are mutually exclusive in this model. When players are in Conflict, they stake their production bundles on the outcome of the Conflict. Those goods are not available to be traded since ownership is ambiguous as trade occurs. Trade requires mutual consent. Until a Conflict is resolved, it isn’t clear who can provide that consent. Therefore, a single player can not be connected via both a Conflict edge and a Trade edge; the Conflict edge takes precedence.

The set of combatants is

$$\mathcal{N}^C = \{n \in \mathcal{N} : \exists i \in \mathcal{N}, e_{in} \in E^C\} \quad (18)$$

And the set of non-combatants is

$$\mathcal{N}^T = \mathcal{N} \setminus \mathcal{N}^C \quad (19)$$

Then, because Conflict takes precedence and Trade requires the consent of both players, a Trade edge requires both players to be in the set of non-combatants and both to adopt a ‘trade’ posture.

$$E^T = \{e_{ij} : a_{i,j} = a_{j,i} = \text{trade} \quad i, j \in \mathcal{N}^T, i < j\} \quad (20)$$

On the resulting interaction graph

$$G = (\mathcal{N}, E^C, E^T) \quad (21)$$

any remaining isolated vertices represent Autarkic players.

$$\mathcal{N}^A = \{n \in \mathcal{N}^T : \nexists i \in \mathcal{N}^T, e_{in} \in E^T\} \quad (22)$$

3.0.7 Conflicts and Markets

I define a *Conflict Structure* $\mathcal{C}$ as the collection of non-trivial connected components of the subgraph $G^C = (\mathcal{N}, E^C)$ of the interaction graph G . Each C_k is referred to as a *Conflict*.

$$\mathcal{C} \equiv \{C_1, \dots, C_K\} \quad (23)$$

Likewise, a *Market Structure* $\mathcal{D}$ is the collection of non-trivial connected components of the subgraph $G^T = (\mathcal{N}, E^T)$ of the interaction graph G . Each D_k is referred to as a *Market*.

$$\mathcal{D} \equiv \{D_1, \dots, D_J\} \quad (24)$$

$V(H)$ denotes the vertex set of graph H , so $V(C_k)$ is the set of players who are combatants in Conflict k .

Likewise, $V(D_k)$ is the set of players trading in Market D_k .

Let H be any connected component of G . Y_H is the aggregate production of the players $V(H)$.

$$Y_H \equiv \sum_{i \in V(H)} X_i^P \quad (25)$$

In a Market D_k , Y_{D_k} is the total supply of the market. In a conflict C_k , Y_{C_k} is the prize of the contest that the players $V(C_k)$ are competing for. For Autarkic player n , $Y_{\{n\}} = X_n^P$, is simply n 's consumer good production.

3.0.8 Coalition Structure

A *coalition structure* $\mathcal{B}$ is a partition of $\mathcal{N}$ that refines the connected-component partition of the interaction graph G .

$$\forall B \in \mathcal{B}, \quad \exists H \in \text{Comp}(G) \text{ such that } B \subseteq V(H) \quad (26)$$

Equation (26) disallows coalitions that span different connected components of the graph. Situations in which one member of a coalition is in Conflict while another trades are therefore not allowed, nor are coalitions whose members participate in different contests or different Markets. This restriction neither imposes a coalition structure prior to the interaction structure nor constructs a coalition structure on a pre-existing interaction structure. Rather, both are induced simultaneously by the same strategy profile: the choices that determine coalition membership also determine the Conflict and Trade structure of the graph. The restriction therefore requires only that the coalition structure be consistent with the interaction structure jointly induced by those choices. Coalition structure and interaction structure are thus jointly determined endogenous objects.

Because the coalition structure permits singletons to operate alone and sets of players to work together, the outcome function takes the coalition as the basic unit when assigning outcomes. However, consumption bundles and payoffs are ultimately assigned to individuals. The outcome function must therefore include a distributional rule that assigns coalition outcomes to individual members⁴.

$$\gamma : B \rightarrow [0, 1], \quad \sum_{i \in B} \gamma_i = 1 \quad (27)$$

I discuss γ further in the results, leaving it unspecified for the moment.

3.0.9 Outcome Function

The outcome function assigns a lottery over coalition consumption bundle allocations to each connected component $H \in \text{Comp}(G)$. These component lotteries are independent, so the joint lottery over the economy-wide consumption allocation is their product. The coalition bundle is subdivided among the players of the coalition using some rule γ .

⁴I consider the distributional rule an institution, although not all economists would agree.

$$X_n^F = \gamma_n X_B^F \quad (28)$$

Because players have common homothetic preferences, they have identical demand schedules, so any such division — regardless of how γ is set — leaves each member's consumption bundle proportional to the coalition's bundle X_B^F , simultaneously satisfying the demand schedule for each player.

An autarkic player is an isolated vertex of G and therefore constitutes a singleton component whose lottery is degenerate at the player's production bundle.

$$X_n^A = X_n^P \quad (29)$$

The assumption of common homothetic preferences is sufficient to guarantee a unique price vector that clears a Market D_k (Chipman 1974; Kehoe 1985; Quah 1993). Because coalitions have identical demand schedules and markets clear, each coalition's consumption bundle is some proportion α_B of aggregate Market production Y_{D_k} . The Market lottery is degenerate also so for player $n \in B$,

$$X_n^T = \gamma_n (\alpha_B Y_{D_k}) \quad (30)$$

For a Conflict C_k , the lottery is not degenerate. The probabilities of winning the contest are determined by a ratio-form contest success function based on the security budgets of each coalition. Because the contest is winner-take-all, the aggregate production of consumption goods by all coalitions in the Conflict, Y_{C_k} , is the prize of the contest.

If W_B is the event that coalition $B \in \mathcal{B}_{C_k}$ wins the contest in Conflict C_k , then q_B is the probability that coalition B wins the contest.

$$q_B \equiv \Pr(W_B) = \frac{\sum_{i \in B} f_{is}(s_i)}{\sum_{B' \in \mathcal{B}_{C_k}} \sum_{j \in B'} f_{js}(s_j)} \quad (31)$$

Then g_{C_k} defines the lottery

$$\{(q_B, Y_{C_k})\}_{B \in \mathcal{B}_{C_k}} \quad (32)$$

Then, finally, if H is a connected component of the interaction graph G , then:

$$g(\sigma) = \bigotimes_{H \in \text{Comp}(G)} g_H(\sigma) \quad (33)$$

3.0.10 Payoff Function

With the partition on $\mathcal{N}$ and the outcome function fully defined, I revisit the payoff function $\pi = v \circ g$. The outcome function g maps strategy profiles to a joint lottery over consumption bundle allocations. The value function v then assigns each player their expected utility of the independent sublottery that they participate in.

$$\pi = v(g(\sigma)) = E_{g(\sigma)} [U(X_i^F)] \quad \forall i \in \mathcal{N} \quad (34)$$

3.1 Equilibrium Concept

I analyze pure-strategy subgame perfect equilibria of the theft and trade game. A characterization of coalition formation requires further restrictions on production and utility functions. To preserve the model's generality, I examine coalition stability against unilateral deviations without establishing coalition-proofness.

A subgame perfect equilibrium is a strategy profile $\sigma^* = (\sigma_1, \sigma_2, \dots, \sigma_N)$ whose restriction to every subgame is a Nash equilibrium of that subgame (Osborne and Rubinstein 2005).

A *Trade equilibrium* is an SPE where all players trade.

Definition 1 Trade Equilibrium: A Trade Equilibrium is a subgame perfect equilibrium in which $E^C(\sigma^*) = \emptyset$

A *Social Order equilibrium*, on the other hand, is an SPE where some players trade. A Social Order equilibrium allows the coexistence of Trade and Conflict.

Definition 2 Social Order Equilibrium: A Social Order equilibrium is a subgame perfect equilibrium in which $E^T(\sigma^*) \neq \emptyset$

Therefore, a Trade equilibrium is a Social Order equilibrium where no Conflict is present. In either case, Autarkic players are excluded by equilibrium behavior, not by definition.

3.2 Remarks

Before turning to equilibrium analysis, I make two implications of the model's treatment of security explicit.

Remark 1: *In equilibrium, each player in a singleton coalition allocates a positive amount of labor to security, $s_n > 0$. A player choosing to forgo security while working alone invites aggression: any other player with a positive amount of security wins a bilateral conflict with certainty and takes both players' production. Forgoing security therefore leaves the unprotected player with zero utility, whereas allocating even a small positive amount of labor to security yields a positive expected payoff regardless of the interaction.*

Remark 2: *There is no market for security goods. Under Conflict, players do not exchange goods. Once engaged in Trade, a player has no remaining use for security, and the labor devoted to its production is a sunk resource cost with no consumption benefit. No market participant would therefore surrender consumer goods in exchange for security. Because Trade and Conflict are mutually exclusive outcomes of the same stage, no market for security forms.*

4 Equilibrium Analysis

4.0.1 Equilibria Characterization and Social Order

Classical trade theory compares the outcomes of Trade with Autarky; the invisible hand of trade allows everyone to optimize jointly which, when compared to individual optimization of production is a near universal improvement. The gains from trade need only outperform a uniformly poor alternative. The theft and trade model replaces the poor alternative with a more attractive one. By allowing every player to declare theft rather than simply withdraw, it provides players with an option where someone is better off rather than making everyone poorer. The Conflict option rewards whoever prevails in the contest, so any option that fails to generate a reward at least as rewarding as the player expects from the contest gets rejected. The discipline this imposes at equilibrium is therefore stronger than the discipline imposed with a comparison to Autarky; a player's payoff no longer merely needs to exceed what she would receive in isolation, because isolation is not the relevant comparison once theft is available. What her payoff must match is what she could obtain by successfully appropriating the economy's output through force.

Characterizing the subgame perfect equilibria of this game is therefore a matter of pinning down exactly what the Conflict alternative delivers to each player. A smaller-scale deviation, stealing from a subset of players, could pay even better than a deviation to universal theft, if that subset happens to be weak relative

to the deviator. But whether such an advantageous subset exists, and which one it is, depends on the rest of the strategy profile. Not every game guarantees that a lucrative subset exists. A declaration of theft against the entire player set, by contrast, is always available, whether the deviator currently trades, fights, or stands alone, so it is the universal theft payoff, not the payoff of withdrawal, that any equilibrium must reproduce. I first show that an SPE exists, and that SPE is the equilibrium that all others must match or improve upon. The following two lemmas are used to show the conditions required to apply Reny's existence result (Reny 1999). They will be used again later to show additional results. The first establishes strict concavity of the contest success function.

Lemma 0.1 Concavity of the Contest Share: Fix $n \in N$ and $S_{-n} > 0$. The contest share

$$q_n(s_n) = \frac{f_{n,S}(s_n)}{f_{n,S}(s_n) + S_{-n}}$$

is strictly concave in s_n .

Proof.

$$q'_n(s_n) = \frac{f'_{n,S}(s_n) S_{-n}}{(f_{n,S}(s_n) + S_{-n})^2},$$

$$q''_n(s_n) = \frac{S_{-n} [f''_{n,S}(s_n)(f_{n,S}(s_n) + S_{-n}) - 2f'_{n,S}(s_n)^2]}{(f_{n,S}(s_n) + S_{-n})^3}.$$

Since $f''_{n,S}(s_n) < 0$, both $f''_{n,S}(s_n)(f_{n,S}(s_n) + S_{-n})$ and $-2f'_{n,S}(s_n)^2$ are strictly negative, so the bracketed term is strictly negative; as $S_{-n} > 0$ and $(f_{n,S}(s_n) + S_{-n})^3 > 0$, $q''_n(s_n) < 0$. $\square$

The second combines the quasiconcavity and homogeneity assumptions on the utility function to show that the utility function is concave.

Lemma 0.2 Concavity of U : U is concave. For x, y with $U(x), U(y) > 0$, the concavity inequality holds with equality if and only if x and y are nonnegative scalar multiples of one another.

Proof. Let x, y have $U(x), U(y) > 0$, and fix $\lambda \in (0, 1)$. Since U is homogeneous of degree one, $U(x/U(x)) = U(y/U(y)) = 1$. By strict quasiconcavity of U , for every $\mu \in (0, 1)$,

$$U\left(\mu \frac{x}{U(x)} + (1 - \mu) \frac{y}{U(y)}\right) \geq 1,$$

with equality only if $x/U(x) = y/U(y)$. Taking $\mu = \frac{\lambda U(x)}{\lambda U(x) + (1 - \lambda)U(y)} \in (0, 1)$ and applying homogeneity

of degree one to clear the denominator gives

$$U(\lambda x + (1 - \lambda)y) \geq \lambda U(x) + (1 - \lambda)U(y),$$

with equality if and only if $x/U(x) = y/U(y)$ — i.e., x and y lie on the same ray from the origin. $\square$

Proposition 1 *Existence of a Battle-Royale SPE:* Any strategy profile $\sigma^* = (L^*, \rho)$ with the following characteristics:

(i) For every $i, j \in N$ with $i \neq j$, and every $L \in \mathcal{L}$: $a_{i,j} = \text{theft}$ or $a_{j,i} = \text{theft}$

(ii) For every $i \in N$, $\pi_i^C(L_i^*, L_{-i}^*) \geq \pi_i^C(L_i, L_{-i}^*)$ for every $L_i \in \mathcal{L}_i$

(iii) For every $i \in N$ such that no $j \neq i$ satisfies $a_{j,i} = \text{theft}$: $\pi_i^C(L_i^*, L_{-i}^*) \geq \max_{t_i: \sum_m t_{im} = \tau} U(X_i^P(t_i))$.

is a subgame-perfect equilibrium of the theft and trade game.

Proof. The feasible labor-allocation set $\mathcal{L}_i$ is nonempty, compact, and convex for every $i \in N$. Hence

$$\mathcal{L} = \prod_{i \in N} \mathcal{L}_i$$

is nonempty, compact, and convex.

Fix $L_{-i} \in \mathcal{L}_{-i}$. By Lemma 0.1, $q_i(s_i)$ is strictly concave in s_i , hence $\ln q_i(s_i)$ is concave in s_i , since a positive concave function is log-concave. By Lemma 0.2, U is concave; since $t_i \mapsto \sum_{j \in N} X_j^P(t_j)$ is concave in t_i (production is strictly concave in t_i , with every other player's contribution fixed), $\ln U(\sum_{j \in N} X_j^P(t_j))$ is concave in t_i , as the composition of the concave nondecreasing function $\ln U$ with a concave function.

$$\ln \pi_i^C(t_i, s_i, L_{-i}) = \ln q_i(s_i) + \ln U\left(\sum_{j \in N} X_j^P(t_j)\right)$$

is therefore a sum of functions each concave in its own block of $L_i = (t_i, s_i)$, hence jointly concave in L_i , and therefore jointly quasiconcave in L_i . Since $\exp$ is strictly increasing, $\pi_i^C(\cdot, L_{-i}) = \exp(\ln \pi_i^C(\cdot, L_{-i}))$ is quasiconcave in L_i as well.

Since $\pi_i^C(\cdot, L_{-i})$ is continuous at every profile with $s \neq \mathbf{0}$, better-reply security holds trivially there: for any non-equilibrium L^0 with $s^0 \neq \mathbf{0}$, continuity gives $v = \pi^C(L^0)$, and since L^0 is not an equilibrium, some player already secures a strictly higher payoff at L^0 itself. It remains only to verify better-reply security at the single profile $s = \mathbf{0}$. Consider any sequence $L^k \rightarrow L^0$ such that $s^k \rightarrow \mathbf{0}$, and suppose that the associated

payoff vectors converge to v . Along a subsequence, the contest probabilities converge to some $q \in \Delta(N)$. Since

$$\sum_{i \in N} q_i = 1,$$

there exists, for $|N| > 1$, some player i such that $q_i < 1$.

Fix $\hat{s}_i > 0$. Since $f_j^S(0) = 0$ for every $j \in N$ and $f_i^S(\hat{s}_i) > 0$,

$$\lim_{s_{-i} \rightarrow \mathbf{0}} q_i(\hat{s}_i, s_{-i}) = 1$$

Moreover, $\hat{s}_i$ may be chosen arbitrarily small. By continuity of production and utility, it can therefore be chosen sufficiently small that the productive cost of $\hat{s}_i$ is smaller than the payoff gain obtained by replacing the limiting contest share $q_i < 1$ with a contest share approaching one. Thus, for an associated feasible productive-labor allocation $\hat{t}_i$,

$$\pi_i^C(\hat{L}_i, L_{-i}^0) > v_i, \quad \hat{L}_i = (\hat{t}_i, \hat{s}_i).$$

Since $\hat{s}_i > 0$, the payoff function is continuous at $(\hat{L}_i, L_{-i}^0)$. Hence there exist $\delta > 0$ and a neighborhood V_{-i} of L_{-i}^0 such that

$$\pi_i^C(\hat{L}_i, L_{-i}) > v_i + \delta \quad \forall L_{-i} \in V_{-i}$$

Player i can therefore secure a payoff strictly greater than v_i , so the Conflict game is better-reply secure.

The Conflict game consequently satisfies the conditions of Reny (1999): strategy sets are nonempty, compact, and convex; payoffs are quasiconcave in each player's own strategy; and the game is better-reply secure. It therefore possesses a pure-strategy Nash equilibrium

$$L^* = (L_i^*)_{i \in N}$$

Following any history, condition (i) guarantees that a unilateral posture change by any player n cannot alter any other player's membership in the Conflict component, since it can only remove edges n herself is party to, and every pair not involving n retains its own independent conflict edge. Nor can a deviation alter n 's own membership, unless n is a player whose theft declarations are the sole source of every one of her edges. Only the complete, simultaneous withdrawal from every pairing removes such a player from Conflict, since any partial withdrawal leaves her connected through her remaining declared edges. Following every history choosing theft is therefore a best response for a player in every pair for all but the most aggressive players, by condition (i) alone; for the most aggressive players, condition (iii) rules out the one alternative condition

(i) does not cover. The unilateral deviation where the most aggressive player chooses Autarky reduces that player's payoff by assumption. The prescribed continuation strategies thus constitute a Nash equilibrium in every continuation subgame.

Given these continuation strategies and condition (i), battle-royale Conflict follows every labor-allocation profile, so the payoffs of the induced Conflict game coincide exactly with the payoffs realized at the labor-allocation stage of the original game. Since L^* is a Nash equilibrium of the induced Conflict game,

$$\pi_i^C(L_i^*, L_{-i}^*) \geq \pi_i^C(L_i, L_{-i}^*) \quad \forall L_i \in \mathcal{L}_i \text{ and } i \in N,$$

this inequality is therefore also the best-response condition for the labor-allocation stage of the original game: no player has a profitable deviation there either. The resulting strategy profile is a Nash equilibrium in every subgame and is therefore a subgame-perfect equilibrium. $\square$

Proposition 1 shows that a battle-royale SPE exists for a theft and trade game. Any player can unilaterally deviate to theft against every other player, triggering a Conflict among all of N ; because that Conflict is itself an equilibrium, the resulting payoff is well-defined rather than a merely hypothetical bound. Because this deviation is available to any player at any strategy profile, the battle-royale payoff is a floor for any other SPE: any other SPE must equal or improve upon it for every player.

Theorem 1 Characterizing SPE in Theft and Trade games: Let $Y_N = \sum_{i \in N} X_i^P$ and σ^* be an SPE. For every $n \in N$,

$$\pi_n(\sigma^*) = \frac{S_n}{\sum_{i \in N} S_i} U(Y_N)$$

The theorem shows that at an SPE, every player's payoff share is exactly their probability of winning in a universal war. Their beginning state, Conflict, Trade or Autarky is irrelevant. Their production capacity is only relevant to the size of the economy, not to their share of production. I prove that this condition is necessary in several steps, and then discuss the implications this theorem has on social order.

The first step in characterizing the necessary conditions for any SPE in a theft and trade game is to show that the labor allocation across consumption goods a player chooses is invariant to the later choice of theft and trade once the security budget is set.

Lemma 1.1 Common Production: Fix security budgets $(s_j)_{j \in V(H)}$ for a set of players $V(H) \subseteq N$ in some non-trivial connected component H of the graph $G(\sigma)$. If H is a Market D_k , let t_n^T denote player n 's optimal production choice at the market-clearing price vector. If instead H is a Conflict C_k , let t_n^C denote

player n 's optimal production choice given Conflict among the set of players $V(H)$. Then $t_n^T = t_n^C$ for every $n \in V(H)$.

Proof. Under both regimes, aggregate production $Y(t_n) = \sum_{i \in V(H)} X_i^P(t_i)$ is the same function of n 's own t_n , holding $(t_i)_{i \neq n}$ and $(s_j)_{j \in V(H)}$ fixed. Under Trade, n maximizes $U(X_n^T) = U(\gamma_n \alpha_B Y(t_n))$; since U is homogeneous of degree one, this equals $\gamma_n \alpha_B U(Y(t_n))$. Under Conflict, n maximizes $\gamma_n q_B U(Y(t_n))$. Both objectives are therefore positive scalar multiples of the same function $U(Y(t_n))$, so they share the same maximizer. Since U is strictly quasiconcave and increasing, and $Y(t_n)$ is strictly concave, $U(Y(t_n))$ is strictly quasiconcave in t_n ; continuity on the compact, convex constraint set gives existence of a maximizer, and strict quasiconcavity gives uniqueness. Hence $t_n^T = t_n^C$. $\square$

The result is a statement about the structure of the market institution and the particular contest form. A Market takes in the goods a player produced and reallocates those goods to the participants in the market. The Conflict contest also reallocates the total sum of goods among the players using a different mechanism. But the same goods are being reallocated. Each version of the reallocation mechanism relies on players behaving as if they were jointly optimizing production over the goods being allocated. Which of the two mechanisms is ultimately employed doesn't matter, since both assign each player a fixed proportional share of the aggregate bundle — a share that does not depend on the player's own production choice.

The result is a statement about the structure of the Market and Conflict institutions. A Market reallocates players' pooled production through the price mechanism that clears markets; the Conflict contest reallocates the same pooled bundle through a lottery with security-weighted shares. A player's privately optimal production choice therefore satisfies exactly the condition a planner maximizing aggregate utility $U(Y)$ over the joint labor profile would impose on her. Hence, production proceeds as if the group were jointly optimizing the aggregate bundle under either mechanism. The Common Production lemma shows that this incentive survives the switch from Trade to Conflict unchanged. The result also sharpens what separates Conflict from Autarky as an outside option: Autarky pools nothing, so no aggregate-maximizing discipline applies to an isolated player's production choice at all. Conflict, despite dividing the pooled bundle adversarially, still shares Trade's cooperative production incentive. Trade and Conflict differ in how the pie is split, not in whether players behave, in production, as if maximizing the size of the pie.

The battle-royale proposition established existence of an SPE for one particular partition of $\mathcal{N}$, the single Conflict spanning every player. Theorem 1, however, must characterize every SPE, and an arbitrary SPE can induce any partition of $\mathcal{N}$ into many Markets, many Conflicts, and Autarkic singletons — not only the battle-royale single Conflict. The following two results show that security weighted shares of economy-wide

production must hold at two levels - the individual level and the component level. Together they let the proof of Theorem 1 sum payoffs across whatever partition an arbitrary equilibrium happens to produce, rather than specific partition of one identified SPE.

The first lemma establishes that an individual's payoff is a proportional share of her coalition's share of the component's aggregate payoff. Furthermore, the coalition's share of the component's aggregate payoff is itself a proportional share.

Lemma 1.2 Proportional Division within a Component: *For any connected component H of $G(\sigma^*)$ and any player $n \in V(H)$ belonging to coalition $B \in \mathcal{B}(\sigma^*)$ with $B \subseteq V(H)$,*

$$\pi_n(\sigma^*) = \gamma_n \omega_B U(Y_H),$$

where $\omega_B = \alpha_B$ if H is a Market, $\omega_B = q_B$ if H is a Conflict, and $\gamma_n = \omega_B = 1$ if $H = \{n\}$ is Autarkic.

Proof. If H is a Market D_k : by (30), $X_n^T = \gamma_n \alpha_B Y_{D_k}$; homogeneity of degree one gives $\pi_n(\sigma^*) = U(X_n^T) = \gamma_n \alpha_B U(Y_{D_k}) = \gamma_n \alpha_B U(Y_H)$.

If H is a Conflict C_k : by (32), $n \in B$ receives $\gamma_n Y_{C_k}$ with probability q_B and 0 otherwise, so $\pi_n(\sigma^*) = \gamma_n q_B U(Y_{C_k}) = \gamma_n q_B U(Y_H)$.

If $H = \{n\}$: by (29), $\pi_n(\sigma^*) = U(Y_{\{n\}}) = U(Y_H)$ trivially. $\square$

Corollary 1.1 Aggregate Payoff Identity: *Summing over $n \in V(H)$, using $\sum_{n \in B} \gamma_n = 1$ and $\sum_B \omega_B = 1$,*

$$\sum_{n \in V(H)} \pi_n(\sigma^*) = U(Y_H).$$

With these three results in hand, Theorem 1 can be proved directly.

Proof of Theorem 1. Fix $n \in \mathcal{N}$. Consider the deviation in which n declares theft toward every other player in round two, regardless of what σ_n^* actually specifies. Since a Conflict edge requires only one side to declare theft (17), and Conflict edges take precedence over Trade edges, this collapses $G(\sigma^*)$ to a single Conflict spanning all of $\mathcal{N}$ — regardless of n 's equilibrium regime (Trade, Conflict, or Autarky) and regardless of the other players' postures toward one another. Production is already sunk when round two begins, so this is a pure continuation deviation.

Since this deviation requires no other player's agreement, n can secure it alone. Setting $B = \{n\}$ in (31) gives her a win probability of exactly $q_n(\sigma^*) = S_n / \sum_{j \in \mathcal{N}} S_j$, since every player is now a combatant in the same

Conflict and the prize is $Y_{\mathcal{N}}$. The deviation's expected payoff to n is therefore $q_n(\sigma^*)U(Y_{\mathcal{N}})$, independent of how the remaining $N - 1$ players would divide the spoils among themselves, since n 's own win probability depends only on the contest's aggregate denominator.

Since σ^* is an SPE, n 's equilibrium payoff must be no worse than this available deviation:

$$\pi_n(\sigma^*) \geq q_n(\sigma^*)U(Y_{\mathcal{N}}) \quad \forall n \in \mathcal{N}. \quad (35)$$

By Corollary 1.1, summing over every component H of $G(\sigma^*)$, which partition $\mathcal{N}$,

$$\sum_{n \in \mathcal{N}} \pi_n(\sigma^*) = \sum_H U(Y_H).$$

Because U is concave (Lemma 0.2) and homogeneous of degree one, it is superadditive: $\sum_H U(Y_H) \leq U(Y_{\mathcal{N}})$, with equality if and only if every Y_H is a nonnegative scalar multiple of $Y_{\mathcal{N}}$. Summing (35) over all n and using $\sum_n q_n(\sigma^*) = 1$,

$$U(Y_{\mathcal{N}}) \leq \sum_{n \in \mathcal{N}} \pi_n(\sigma^*) = \sum_H U(Y_H) \leq U(Y_{\mathcal{N}}).$$

Both inequalities must therefore bind. Equality in the second forces $\sum_n \pi_n(\sigma^*) = \sum_n q_n(\sigma^*)U(Y_{\mathcal{N}})$; since (35) already gives $\pi_n(\sigma^*) \geq q_n(\sigma^*)U(Y_{\mathcal{N}})$ term by term, equality of the sums forces equality in every term:

$$\pi_n(\sigma^*) = q_n(\sigma^*)U(Y_{\mathcal{N}}) \quad \forall n \in \mathcal{N}.$$

□

The impact of the theorem is that every SPE in the theft and trade game will reward players exactly according to their optimum power level. No option, not Trade, not Autarky will improve upon the security weighted share allocation.

Corollary 1.2 Social Order Equilibria: *If σ^* is a Social Order equilibrium ($E^T(\sigma^*) \neq \emptyset$) or a Trade equilibrium ($E^C(\sigma^*) = \emptyset$), then $\pi_n(\sigma^*) = q_n(\sigma^*)U(Y_{\mathcal{N}})$ for every $n \in \mathcal{N}$.*

Proof. Immediate from Theorem 1, which holds for every $n \in \mathcal{N}$ regardless of regime. □

Here is where the distinction between Autarky and Conflict as outside options is felt. Autarky permits a player to disengage completely, but at the cost of foregoing the gains generated by specialization. Conflict is the outside option that binds. A player can claim a share of aggregate production that is determined

by relative coercive power rather than a claim that is determined by their productive input to aggregate production. Skaperdas and Syropoulos (1997) establish the proportionality principle in an economy in which agents individually produce coercive power but jointly produce a single consumption good. The common product is divided according to relative coercive power, either through conflict or an equivalent peaceful division. Here, agents independently produce both coercive power and multiple consumption goods. Joint production is therefore not a technological assumption; Autarky remains feasible. Nevertheless, equilibrium production choices make independently produced goods behave as a common surplus. Autarky provides the option to leave that surplus behind, while Conflict provides the meaningful alternative for claiming it. It is the nature of Conflict that it requires no other player's consent that makes coercive power the scale by which goods are ultimately distributed.

Cox, North, and Weingast (2019) name the proportionality principle and identify it as a necessary condition among the elite organizations that constitute the dominant coalition. The result here extends beyond elites and beyond organizations. The proportionality principle arises from the outside options of individual agents before organizations exist and continues to apply as those agents organize. It is therefore not merely a condition governing the distribution of rents among powerful elites, but an equilibrium restriction on the distribution of economic value wherever coercive alternatives remain available.

Another result that follows immediately from security weighted shares requires the aggregate payoff for each connected component of the graph to also be weighted by the security good production of its members. Although, in equilibrium, Markets may fragment and independent Conflicts may emerge, the fragmentation doesn't carry economic weight. Because all interactions are measured by the same yardstick, differential pricing in separated markets cannot survive. The same logic rules out more lucrative and less profitable sub-Conflicts: since every player's payoff is pinned to the same economy-wide rate $q_n(\sigma^*)U(Y_N)$ regardless of which component she occupies, no Conflict can pay its combatants more, or less, per unit of power than any other. A component that paid less would leave its participants a profitable universal deviation; a component that paid more is exactly what Corollary 1.3's equality condition rules out. Fragmentation of the interaction graph into separate Markets and Conflicts carries no independent equilibrium content — it is the same underlying allocation wearing different graph clothing.

Corollary 1.3 *Structural Irrelevance:* *For any SPE σ^* and any component H of $G(\sigma^*)$, Y_H is a non-negative scalar multiple of Y_N : every component's production is allocated in exactly the same proportions as whole-economy production.*

Proof. This is the equality condition of the superadditivity step in the proof of Theorem 1: both inequalities

bind only when every Y_H is proportional to Y_N . □

A final consequence of the proportionality principle eliminates Autarky as an equilibrium choice except in perverse scenarios. Each player in hypothetical Autarky receives a share of the global economy that is weighted by their relative power. However the Autarkic player hasn't fully contributed to aggregate production so long as there are any gains from specialization remaining across her labor allocation. Therefore, she has a unilateral deviation that increases global production without altering the power structure.

Corollary 1.4 Autarky Exclusion: *Let σ^* be an SPE and let $n \in \mathcal{N}$ be an Autarkic player. Then, for every pair of consumption goods m, m' ,*

$$\frac{f'_{n,m}(t_{n,m})}{f'_{n,m'}(t_{n,m'})} = \frac{\partial U(Y_N)/\partial X_m}{\partial U(Y_N)/\partial X_{m'}},$$

evaluated at n 's equilibrium labor allocation t_n . Autarky can therefore be part of an SPE only if n 's own marginal rate of transformation between every pair of goods already equals the marginal rate of substitution implied by the aggregate bundle Y_N — that is, only if no gains from specialization remain for n relative to the rest of the economy.

Proof. Since n is Autarkic, her consumption bundle equals her production bundle, $X_n^F = X_n^P(t_n)$, and she chooses t_n to maximize $U(X_n^P(t_n))$ subject to her own labor-budget constraint alone, with no reference to any price or to any other player. At an interior optimum, her first-order condition equates, for every pair of goods m, m' ,

$$\frac{\partial U(X_n^P(t_n))/\partial X_m}{\partial U(X_n^P(t_n))/\partial X_{m'}} = \frac{f'_{n,m}(t_{n,m})}{f'_{n,m'}(t_{n,m'})}, \quad (36)$$

her own marginal rate of substitution equal to her own marginal rate of transformation, both evaluated at her own bundle.

By Corollary 1.3, $X_n^P(t_n)$ is a nonnegative scalar multiple of Y_N . Since U is homogeneous of degree one, $\partial U/\partial X_m$ is homogeneous of degree zero, so the ratio $\partial U/\partial X_m/\partial U/\partial X_{m'}$ takes the same value at any two points on the same ray — in particular, the same value at $X_n^P(t_n)$ as at Y_N . Substituting into (36) gives the stated identity.

Since t_n is chosen to maximize n 's own utility using only her own technology $f_{n,m}$, with no reference to Y_N or to any other player's production, nothing in her optimization problem forces her own marginal rate

of transformation to equal the aggregate economy's marginal rate of substitution. The identity holds only when it does so by coincidence. $\square$

In this section, I have shown that a theft and trade game admits a well-defined subgame-perfect equilibrium despite the payoff discontinuity at zero security, and that the equilibrium is non-degenerate: every player holds positive coercive power. Moreover, every SPE must reproduce the payoff allocation derived from the battle-royale SPE. The resulting power-weighted distribution is therefore not an artifact of the battle-royale construction. The value of economic production must be distributed according to relative coercive power in every SPE, including equilibria that admit Trade. This derives the proportionality principle from the equilibrium behavior of individual agents and extends the settings in which Skaperdas and Syropoulos and Cox, North, and Weingast previously identified it.

Equilibrium choices do not preclude Social Order. Players may still choose the Market institution, with its associated respect for property rights. But the game contains no mechanism that aligns the distribution generated by the Market with the distribution of coercive power. Absent coercion, markets generally reward economic productivity rather than coercive power, and nothing in the primitives requires the market allocation and the coercive allocation to coincide. Autarky remains feasible but never binds: it permits a player to disengage completely, while Conflict provides the outside option that disciplines the distribution of economic product.

Much of the familiar economic structure nevertheless survives. The law of one price is maintained and extended. Within Conflict, relative coercive power acts as a price, and the price of consumption goods in power terms must equalize across disconnected Markets and Conflicts. The invisible hand thus makes an appearance in both: players behave as if they were jointly optimizing production regardless of their ultimate interaction with other players in the economy. Providing players with an attractive outside option to Trade in the form of Conflict is therefore not innocuous. Trade itself becomes more difficult to sustain, even while much of the economic scaffolding ordinarily associated with Trade remains intact.

5 Organizations

Theorem 1 treats every player as though she stood alone against the rest of the economy. The door to organizing was deliberately left open in the model construction. Across Markets and Conflicts, players behave as if they are jointly optimizing the size of the pie that they are competing over, but maintain their security individually. Organizing extends the fiction of joint optimization to actual joint optimization that

spans across security also, creating the possibility of expanding the production possibilities frontier of the economy relative to individual optimization. That expansion may not be costless, however. Forming a coalition B that plans production and security jointly requires players to agree on a division of the surplus they generate. The distribution rule that allocates the payoffs within the coalition may itself play a decisive role regarding which coalitions form at equilibrium.

Therefore, organizing raises a natural question: can joint optimization create gains that coalition members cannot secure individually, and if so, can those gains be divided in a way that sustains the coalition? Joint optimization determines the gains available to the coalition; the distribution rule γ determines whether its members can be induced to realize them. The two natural candidates for γ trade off efficiency against enforceability in opposite directions, and the enforceability cost is the classic rent-dissipation problem of group contests (Münster 2009; Nitzan 1991; Skaperdas 1996).

Lemma 1.3 *Equal Shares Induce a Team Game:* *Fix a coalition $B \subseteq V(C_k)$ with distribution rule $\gamma_i = 1/|B|$ for every $i \in B$. Then a Nash equilibrium of the induced game among members of B — production and security choices, holding every other player’s strategy fixed — coincides with the joint-surplus-maximizing allocation $\arg \max_{(L_i)_{i \in B}} q_B(S_B) U(Y_{C_k})$.*

Proof. For every $i \in B$, $\pi_i = \gamma_i q_B(S_B) U(Y_{C_k}) = \frac{1}{|B|} q_B(S_B) U(Y_{C_k})$. Because $\frac{1}{|B|}$ is the same positive constant for every member and does not depend on any player’s own choice of L_i , i ’s best response solves $\max_{L_i} q_B(S_B) U(Y_{C_k})$ — the identical problem for every member. A Nash equilibrium L_B^* of the induced game therefore satisfies $\partial [q_B(S_B) U(Y_{C_k})] / \partial L_i(L_B^*) = 0$ for every $i \in B$ simultaneously, which is exactly the condition for L_B^* to be a stationary point of $q_B(S_B) U(Y_{C_k})$ over the full joint vector $(L_i)_{i \in B}$. Since $q_B(S_B) U(Y_{C_k})$ is strictly quasiconcave over this joint vector (Lemmas 0.1 and 0.2, applied to B in place of a single player and extended coordinate-wise across B ’s members), a stationary point is its unique global maximizer. Hence $L_B^* = \arg \max_{(L_i)_{i \in B}} q_B(S_B) U(Y_{C_k})$. $\square$

Equal shares achieve efficiency by construction; every member is maximizing the same objective. However, efficiency is not the only requirement a coalition must satisfy in this model. Each member of the coalition makes her own choices between cooperation and appropriation. If the coalition doesn’t survive its members’ own continuation choices, then the coalition does not form in equilibrium.

Proposition 2 *Equal Shares Are Not Self-Enforcing:* *Fix a coalition B with equal shares $\gamma_i = 1/w$. If any member $i \in B$ has $S_i/S_B > 1/w$ — her share of the coalition’s own security exceeds her assigned equal share of its payoff — then i has a profitable unilateral deviation: declaring theft against every other member*

of B converts B 's continuation into a Conflict among its own former members, in which i 's win probability is exactly S_i/S_B , strictly exceeding her equal share $1/w$ of the same prize.

Proof. Declaring theft toward every $j \in B \setminus \{i\}$ merges B into a single Conflict, by the same graph mechanics used in the proof of Theorem 1, with prize Y_B and i 's win probability $S_i/\sum_{j \in B} S_j = S_i/S_B$. This yields i a payoff of $(S_i/S_B)U(Y_B)$, strictly exceeding her equal share $\frac{1}{w}U(Y_B)$ whenever $S_i/S_B > 1/w$. $\square$

An equal-share coalition is therefore only ever as stable as its members' relative power happens to be balanced — a coincidence the model gives no reason to expect. The alternative is to tie each member's share continuously to her own relative power.

5.0.1 Player Optimization

In the first round of the game, every player is allocating their labor to optimize their eventual payoff. In round two, the player choices determine a partition over the set of players. The partition describes regions where the players may be trading in a market or fighting over their goods. In some regions, player n may have been dragged into Conflict by another player's strategy. In other regions, player n may have the choice herself over whether the players are in Conflict or a Market. There is, in each region, however, the possibility of a Conflict.

In determining the optimal payoff, then, the player's task is to optimize over each region, and, insofar as the player has an option, choose the interaction of the region that maximizes her payoff. In other words, player n 's has a piecewise objective function over the potential regions defined by the other players' strategies and her own potential responses.

Lemma 2.1 Regional Security Optimization: *Fix a region R_k of $\mathcal{L}_n$ induced by sigma_{-n} , with win probability $q_n(s_n)$ given by (31) for the singleton coalition $\{n\}$, where $S_{-n} \equiv \sum_{j \neq n} S_j$ is fixed. Player n 's optimal security budget $s_n^{(k)}$ satisfies, for every consumption good m ,*

$$1 - q_n(s_n^{(k)}) = \frac{\frac{d}{dt_{n,m}} \ln U(Y_{C_k})}{\frac{d}{ds_n} \ln f_{n,S}(s_n^{(k)})},$$

and this equation has a unique solution.

This characterization is not special to players whose equilibrium regime turns out to be Conflict. Security is chosen in round one, before round two's posture choices settle which regime any player ends up in, and s_n

affects her payoff only through how it would perform in whatever Conflict she might be drawn into — by her own declaration or another's. Region R_k is therefore one branch of the same piecewise optimization every player evaluates, whether or not it is the branch she ultimately occupies; a player who ends up in Trade or Autarky still sets s_n by weighing this same margin against the alternative of appropriation.

The probability that n loses the contest equals the ratio of two logarithmic derivatives, both taken with respect to a unit of her own labor: the marginal return, in utility, to allocating that unit toward producing good m , relative to the marginal return to allocating it toward security instead. Because the left side does not depend on m , this ratio is equalized across every consumption good n produces — exactly as a competitive factor market equalizes a mobile input's value of marginal product across its uses.

Proof. For every consumption good m , the Lagrangian first-order conditions on the labor constraint give

$$q_n(s_n) U'(Y_{C_k}) f'_{n,m}(t_{n,m}) = \lambda_n = q'_n(s_n) U(Y_{C_k}).$$

Dividing through by $q_n(s_n) U(Y_{C_k})$ and using $\frac{dY_{C_k}}{dt_{n,m}} = f'_{n,m}(t_{n,m})$ to write the left side as a single logarithmic derivative with respect to $t_{n,m}$,

$$\frac{d}{dt_{n,m}} \ln U(Y_{C_k}) = \frac{U'(Y_{C_k})}{U(Y_{C_k})} f'_{n,m}(t_{n,m}) = \frac{q'_n(s_n)}{q_n(s_n)}.$$

Since $q_n(s_n) = f_{n,S}(s_n)/(f_{n,S}(s_n) + S_{-n})$ with S_{-n} fixed,

$$\frac{q'_n(s_n)}{q_n(s_n)} = \frac{f'_{n,S}(s_n)}{f_{n,S}(s_n)} (1 - q_n(s_n)) = \frac{d}{ds_n} \ln f_{n,S}(s_n) \cdot (1 - q_n(s_n))$$

for any s_n . Combining the two displayed identities and solving for $1 - q_n(s_n)$ gives the equation in the lemma.

$V_n(t_n, s_n)$ is strictly concave on the labor-budget constraint set: $f_{n,m}$ and $f_{n,S}$ are each C^2 , strictly increasing, and strictly concave by assumption, and the ratio-form $q_n(s_n)$ is itself strictly concave in s_n for any such $f_{n,S}$ (Corchón 2007; Cornes and Hartley 2005; Skaperdas 1996). Continuity of V_n on the compact constraint set gives existence of a maximizer; strict concavity gives uniqueness. The displayed equation is therefore the first-order condition of the unique argmax, not merely a stationary point among several. $\square$

Lemma 2.2 Security Budget Existence: $\mathcal{N}$ is finite, so only finitely many graphs and coalition structures can be induced by ρ_{-n} : $\mathcal{L}_n$ partitions into finitely many regions $R_1, \dots, R_K$. By Lemma 2.2, each R_k has a

well-defined value $V_n(t_n^{(k)}, s_n^{(k)})$. Player n 's security budget s_n^* is

$$s_n^* = s_n^{(k^*)}, \quad k^* \in \arg \max_{k \in \{1, \dots, K\}} V_n(t_n^{(k)}, s_n^{(k)}),$$

and k^* exists, since it is the maximizer of finitely many real numbers.

Proposition 3 Telescoping under Nested Proportional Sharing: Suppose coalition $B \subseteq V(C_k)$ uses proportional shares $\gamma_i = S_i/S_B$ for every $i \in B$, and B 's own share of Conflict C_k is itself proportional, $q_B = S_B/S_{C_k}$. Then for every $i \in B$,

$$\pi_i(\sigma^*) = \gamma_i q_B U(Y_{C_k}) = \frac{S_i}{S_{C_k}} U(Y_{C_k}),$$

exactly the payoff i would receive as a lone combatant in C_k . Coalition membership under proportional sharing is payoff-irrelevant: it changes neither i 's equilibrium payoff nor, since her objective is unchanged, her optimal production and security choices.

Proof. Substitute $\gamma_i = S_i/S_B$ and $q_B = S_B/S_{C_k}$ into $\pi_i = \gamma_i q_B U(Y_{C_k})$ (Lemma 1.2); the factor S_B cancels, leaving $\pi_i = (S_i/S_{C_k})U(Y_{C_k})$, which coincides with the payoff a lone combatant would receive by Theorem 1 applied within C_k . Since i 's payoff as a function of her own (t_i, s_i) is identical to the lone-combatant objective, so is her optimal choice. $\square$

Proportional sharing is self-enforcing for exactly the reason equal sharing is not — no member's share ever falls short of what her own relative power would win her — but the telescoping identity shows this comes at a cost: coalition membership accomplishes nothing for its members' payoffs. It also reveals, as a corollary, why proportional sharing is inefficient even on its own terms.

Corollary 3.1 Proportional Sharing Over-Invests in Security: The aggregate security $S_B = \sum_{i \in B} S_i$ that emerges when B 's members individually optimize under proportional sharing equals the aggregate security $|B|$ mutually non-cooperating individuals would choose, which exceeds the security level that maximizes B 's joint payoff $\Phi_B = q_B(S_B)U(Y_{C_k})$ treated as a function of S_B alone (Münster 2009; Nitzan 1991; Skaperdas 1996). The distortion is a pure redistributive artifact of dividing gains by relative power, with no counterpart in Φ_B , and it does not depend on the curvature of $f_{i,S}$.

Proof. By Proposition 3, each $i \in B$'s objective under proportional sharing is identical to the lone-combatant objective, so B 's members choose security exactly as $|B|$ independent combatants would. The joint-optimal choice of S_B instead solves $\max_{S_B} q_B(S_B)U(Y_{C_k})$ directly, internalizing that reallocating security among

B 's own members leaves Φ_B unchanged, since $\sum_{i \in B} \gamma_i = 1$ regardless of how security is distributed within B . Each individual's first-order condition includes a private marginal return from raising her own share of B 's prize at other members' expense, a term with no counterpart in $\partial \Phi_B / \partial S_B$ because that internal redistribution nets to zero when summed across B . This is the rent-dissipation result of Nitzan (1991), generalized to arbitrary increasing $f_{i,S}$ by the ratio-form axiomatics of Skaperdas (1996) and the group-contest generalization of Münster (2009); the distortion does not depend on $f''_{i,S}$, since $f'_{i,S}$ itself cancels from the private-versus-joint marginal comparison (Marschak and Radner 1972). $\square$

Organizing therefore does not fail outright under either rule — coalitions form, and members receive a well-defined share of something. But neither rule realizes the gains specialization within the coalition could create: equal shares are efficient only until a member's relative power outgrows her assigned portion, and proportional shares are stable only by collapsing each member's payoff back to what she would have received acting entirely alone. One coalition remains to be examined — the grand coalition spanning the entire economy, which faces no external rival at all — and it is there that organization and the paper's third headline result turn out to be the same thing.

6 Transfers

Corollary 3.2 *Transfer Schedule Exists:* *Let $\alpha_n(\sigma)$ denote player n 's payoff share under ordinary market clearing, absent any transfer. Define*

$$r_n(\sigma) \equiv (q_n(\sigma) - \alpha_n(\sigma)) U(Y_N).$$

If every player's post-transfer payoff is $\alpha_n(\sigma)U(Y_N) + r_n(\sigma) = q_n(\sigma)U(Y_N)$, then σ satisfies the necessary condition of Theorem 1 for every n , and no player has a profitable deviation to theft: Trade is sustained.

Proof. By construction, the post-transfer payoff to n is exactly $q_n(\sigma)U(Y_N)$, the value of n 's outside option established in the proof of Theorem 1; n is therefore indifferent between transferred Trade and theft, and strictly worse off from any other deviation. Feasibility follows from $\sum_n r_n(\sigma) = (\sum_n q_n(\sigma) - \sum_n \alpha_n(\sigma))U(Y_N) = (1 - 1)U(Y_N) = 0$, since both $\sum_n q_n(\sigma) = 1$ and $\sum_n \alpha_n(\sigma) = 1$: the schedule redistributes among players without requiring resources from outside the economy. $\square$

Proposition 4 *No Peace Dividend:* *Under the transfer schedule of Corollary 3.2, every player's payoff is the identical function $q_n(\sigma)U(Y_N)$ of her own security choice whether she anticipates Conflict or*

successfully-transferred Trade. Her round-one best response s_n^* is therefore unchanged by the presence of transfers: aggregate security production, and the deadweight loss it represents, is the same whether or not transfers succeed in averting violence. Only the physical enactment of conflict is avoided; the resource cost of maintaining relative power is not.

Proof. Since $\pi_n = q_n(\sigma)U(Y_N)$ is the same function of (t_n, s_n) under both continuations by Corollary 3.2, n 's round-one optimization problem, and hence her optimal s_n^* , is identical in either case. $\square$

This is not a coincidence of construction; it is the grand coalition of the previous section, applied to the whole economy.

Theorem 2 Grand Coalition Instability: *Let $B = N$ with equal shares $\gamma_n = 1/N$. By Lemma 1.3, the joint-surplus-maximizing security profile is $s_n = 0$ for every n , since with no external rival $\partial\Phi_N/\partial s_n = U'(Y_N)\partial Y_N/\partial s_n < 0$ for any $s_n > 0$: the grand coalition's own interest is served purely by diverting all labor to production, reaching the Ricardian efficient frontier. This profile cannot be part of any SPE. If every player adopts $s_n = 0$, then $S_n = f_{n,S}(0) = 0$ for every n , so any single player who instead chooses $s_n = \varepsilon > 0$ obtains $S_n > 0 = \sum_{j \neq n} S_j$ and wins the resulting universal conflict with certainty, capturing a payoff approaching $U(Y_N)$ as $\varepsilon \rightarrow 0$ — strictly more than her equal share $\frac{1}{N}U(Y_N)$ for any $N > 1$. The efficient allocation does not merely fail to be stable at the margin; it invites certain, near-costless conquest.*

Proof. Immediate from Remark 1 and Theorem 1, applied to the deviation $s_n = \varepsilon$ against the profile $(s_j = 0)_{j \neq n}$. $\square$

Remark 3 Transfers Are the Grand Coalition in Disguise: *The transfer schedule of Corollary 3.2, applied economy-wide, is exactly the equal-share grand coalition of Theorem 2: aligning every player's payoff with $q_n(\sigma)U(Y_N)$ while realizing the full efficient frontier is precisely what an equal-share grand coalition would achieve, and it inherits the same instability. Any player can abandon the transferred arrangement for a cheap declaration of theft and, since Proposition 4 shows security levels are unaffected by transfers, capture more than her transferred share by exactly the argument of Theorem 2. The schedule establishes the feasibility of economy-wide peace, not its implementation.*

7 Discussion and Conclusion

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